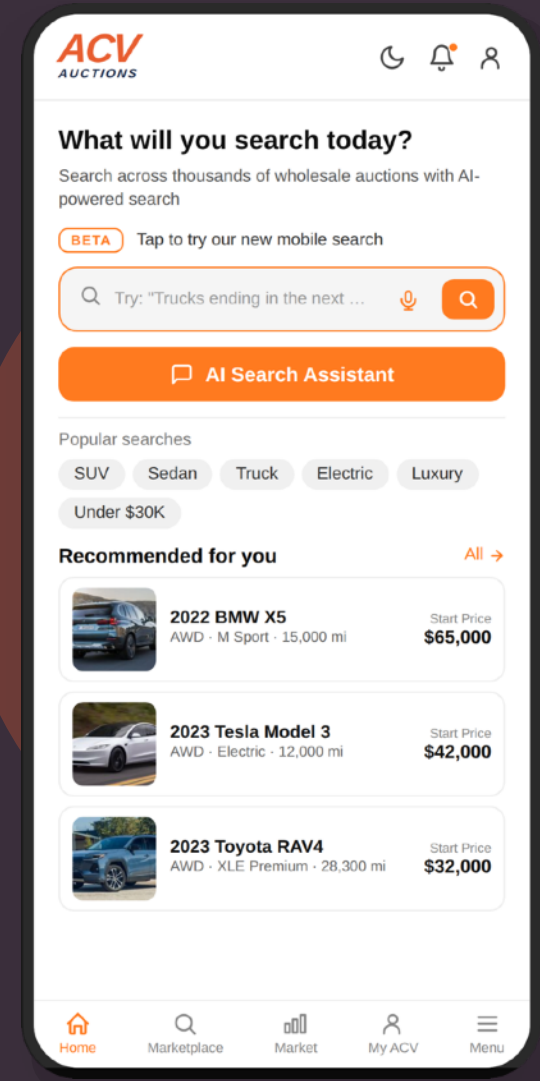


PRODUCT DESIGN CASE STUDY

ACV EASE

AI-powered search & discovery for
wholesale vehicle auctions.

Margaux Kan Senior Product Design Manager



TITLE & OUTCOME

Reimagining how dealers find their next car

ACV EASE is a mobile-first search experience for ACV's wholesale auction marketplace. Instead of stacking rigid filters, buyers can ask for what they want in plain language, by voice or text, and an AI layer surfaces the strongest opportunities, explains why they matter, and guides each buyer from search to bid.



My role

Lead product designer
Research · IxD · UI · prototyping



Timeline

[Add duration]
e.g. 12 weeks, 2025



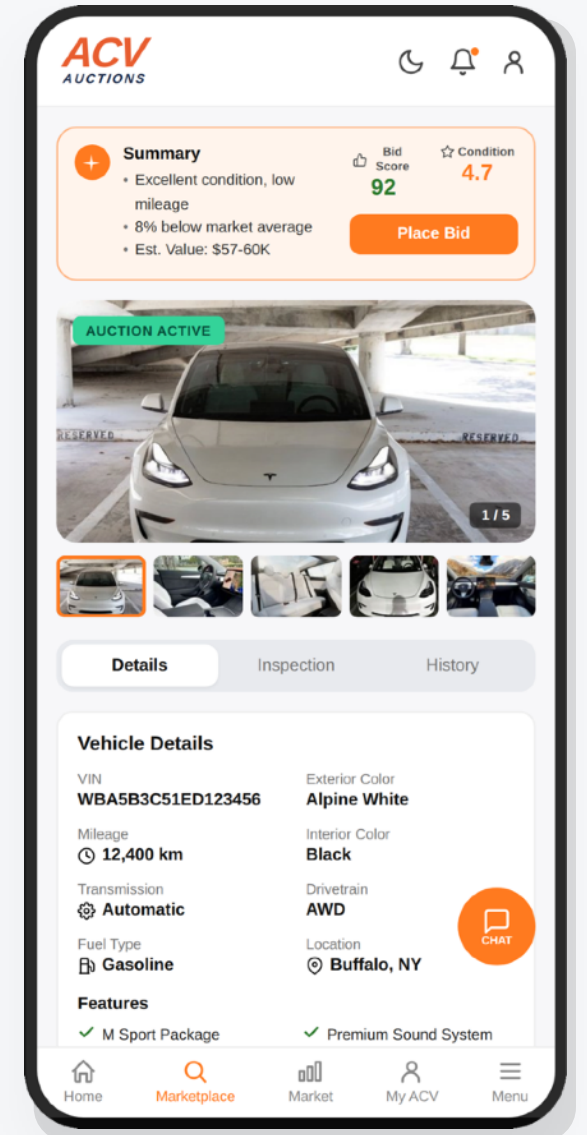
Platform

iOS & Android
ACV mobile app



Team

[Add collaborators]
PM · Eng · Data Science



Finding the right car was slow, manual, and overwhelming

Buyers run a high-volume business from their phones, between appraisals, on the lot, on the move. The old search made them work for it.



Filters, not language

Buyers had to translate what they wanted into rigid drop-downs and checkboxes, and often missed inventory that didn't match the exact filter.



Thousands of live auctions

Relevant vehicles were scattered across constantly-changing auctions, hard to scan on a small screen with seconds to spare.



Good deals stayed buried

Nothing told a buyer which listing was actually a strong opportunity, so the best cars were easy to scroll right past.

Leading search from a strategic bet to launch

Lead product designer: research, interaction design, UI, and prototyping across the search-to-bid journey.

What I led

- Framed AI-powered search as a growth bet, not a UI refresh.
- Defined the vision for conversational, explainable discovery.
- Aligned Product, Engineering, and Data Science around buyer intent.
- Owned the search experience from discovery through beta launch.

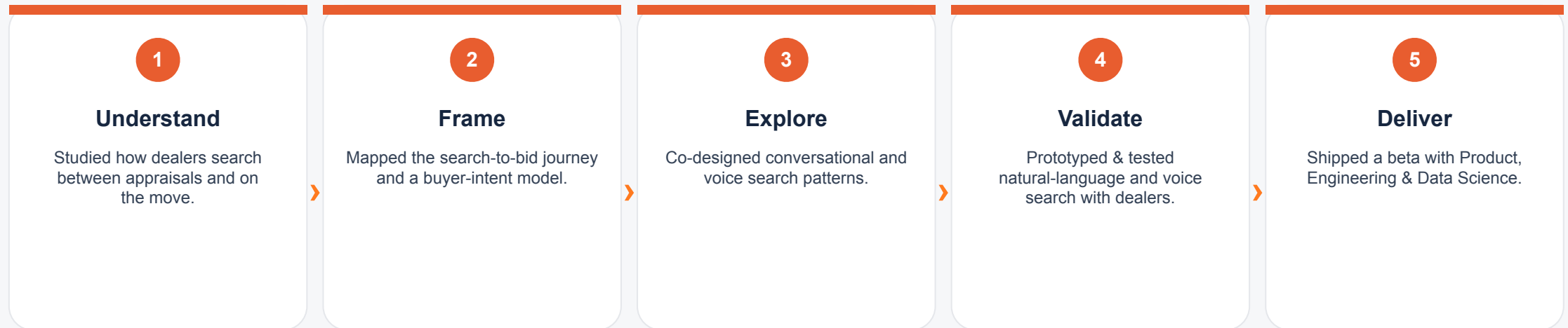
How I led

- Championed a mobile-first experience built for one hand on the lot.
- Partnered with Data Science on ranking and plain-language reasons.
- Led 3 design reviews; balanced speed against buyer trust.
- Used field evidence to resolve trade-offs across teams.

DESIGN PROCESS

A structured path from intent to confident bid

Five phases: pushing design upstream to shape the search strategy, not just the UI.



Leadership lens

Embedded design early to influence the ranking and intent strategy, not just the screens.

Grounding AI search in how buyers actually search

Research led by me, with Product and Data Science embedded as partners.

Methods

- Interviews & ride-alongs with 10 dealers across the U.S.
- Observed live search sessions on the legacy filter experience.
- Audited the old search; usability-tested NL & voice prototypes.

How research was run

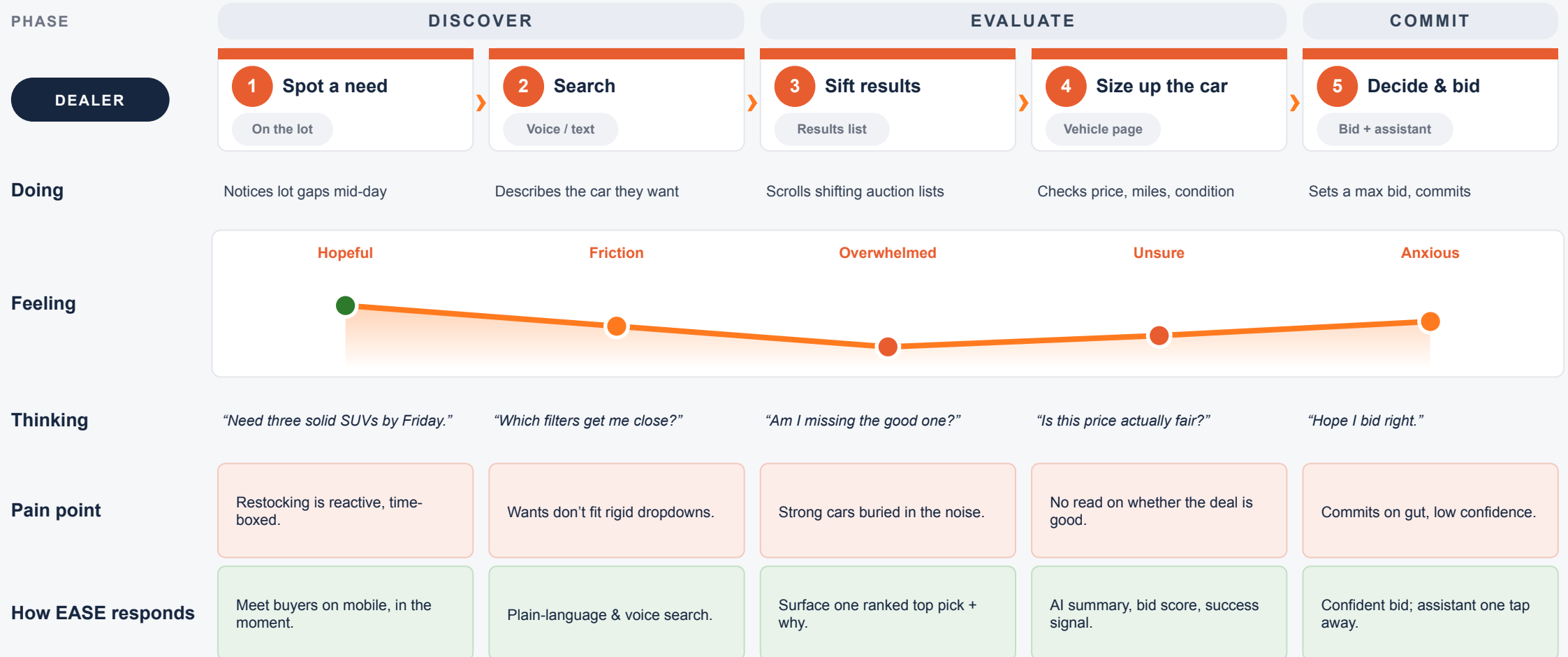
- Research plan led and owned by me.
- PMs embedded as co-researchers; Data Science on intent signals.
- Findings shared across Product, Engineering & Data Science.

Goal Learn where filter search broke down, and what a “good result” means to a buyer with seconds to decide.

USER JOURNEY

From first need to confident bid

The dealer's end-to-end experience: actions, mindset, emotion, pain, and where EASE intervenes.



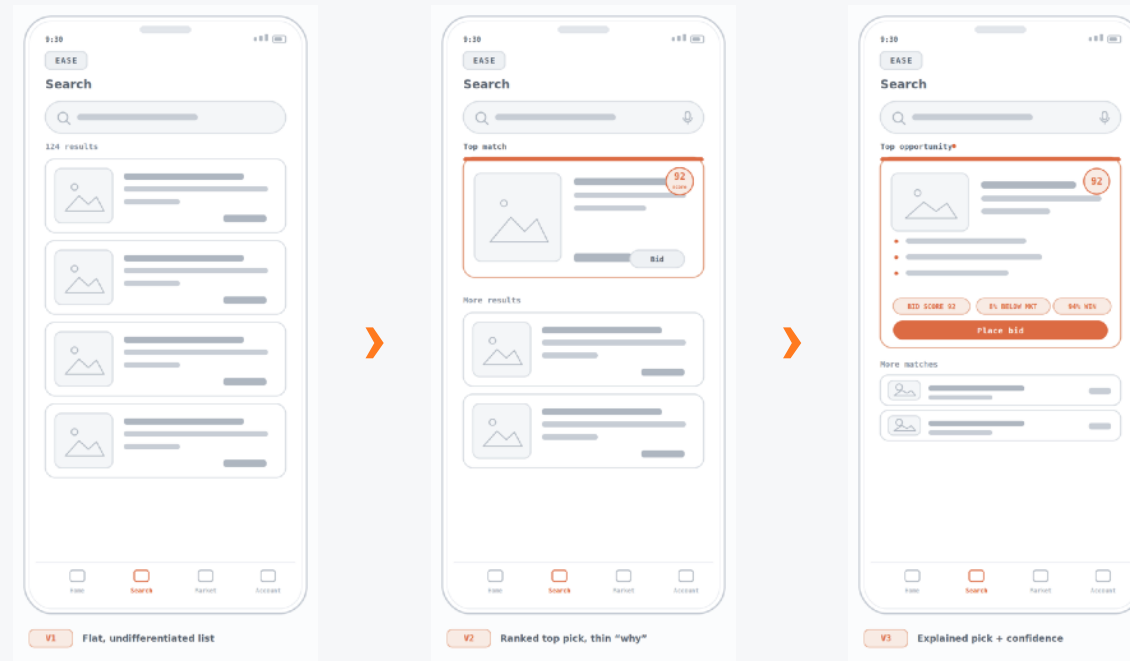
What we heard, and how the design evolved

Three insights reshaped the experience from a list into a recommendation.

Buyers think in plain language, not filter taxonomies.

One strong, explained pick beats a long list of options.

Trust comes from the why: reasons plus a confidence signal.



Key learning Leading with one explained opportunity, plus plain-language and voice input, cut search friction without hiding inventory.

From filtering a catalog to surfacing the best opportunity

Primary Persona: High-Volume Dealer

- Mobile-first; buys between appraisals and on the lot.
- Optimizes for speed, margin, and the next profitable car.
- Low tolerance for friction; high need for trust.

Strategic reframe

We shifted from helping buyers filter inventory to recommending the strongest opportunity, and explaining why it's worth a bid.

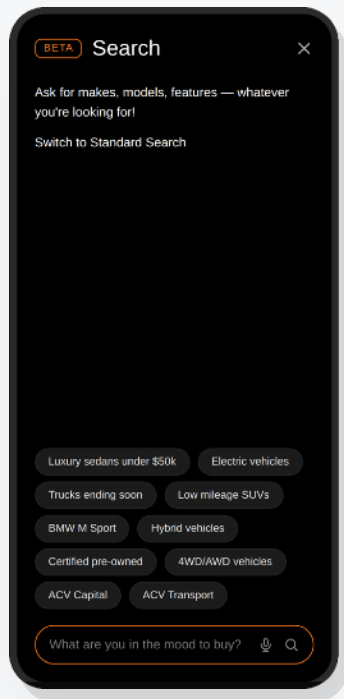
“[Add a dealer or stakeholder quote about EASE here]” [Name, role]

THE SOLUTION

▶ View animated demo

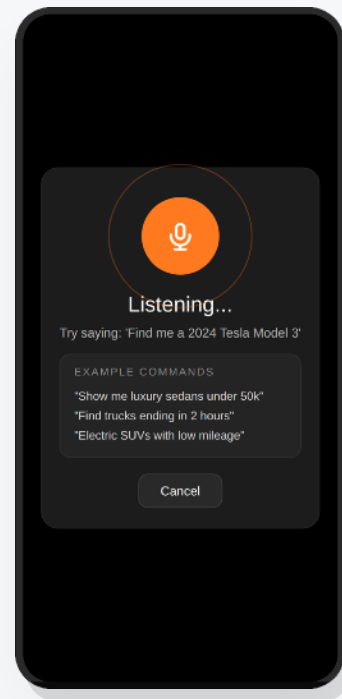
One intelligent flow, end to end

From a spoken request to a confident bid, the experience stays in one continuous, guided thread.



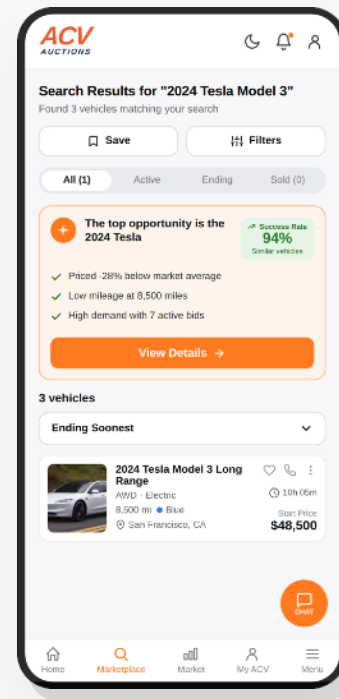
1. Ask

Type or speak a request in plain language.



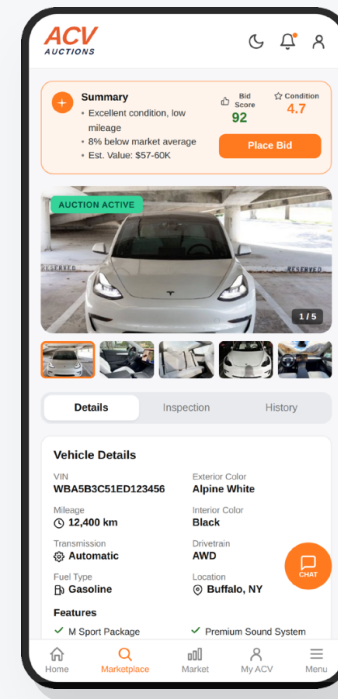
2. Understand

Voice is captured and interpreted instantly.



3. Surface

The top opportunity is ranked and explained.



4. Decide

A focused detail page leads to the bid.

IMPACT

Early signals

Placeholder metrics: swap in your measured results from beta and launch.

↓ **40%**

**Time to first
relevant result**

vs. the legacy filter search

62%

**of beta searches used
voice or natural language**

+18 pts

**Lift in search →
vehicle-detail view**

4.6 / 5

**Beta buyer satisfaction
with the new search**